

b) It defines the maximum length of each type of instruction.

* Since the MIPS is of 32-bits ISA. So, the instruction should be accommodated with that 32-bits.

c) It defines the instruction format of each type of instruction, which determines how the entire instruction is encoded within that 32-bits.

Instruction cycle : —

For an instruction to be executed two registers are used.

i) MAR (Memory address register)

ii) MDR

iii) MBR

It is connected to the data line of the system bus.

There are two mandatory registers i.e. PC and IR which helps to phase the values of operand and executing the instruction.

There are 3 main steps involved in instruction cycle.

1) Fetch

2) Decode

3) Execute

Another 3 steps also involved to complete one instruction cycle.

a) Fetch operands

b) Store the result

c) Interrupt handling